

Laxmi Charitable Trust's Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce Department of Information Technology
(B.Sc.I.T Semester III)

Python For Data Science Practical

Practical - VI

6a. Write a Python program to import a CSV dataset using Pandas and display the first 5 and last 5 records **bold text**

```
import pandas as pd
```

```
df = pd.read_csv("Students.csv")
```

```
print(df.head())
```

	Roll No	Name	Department	Marks	Attendance
0	101	DIVYESH	BCA	78.0	99.0
1	102	DHARMI	BTECH	80.0	90.0
2	103	HEMISH	BCA	88.0	92.0
3	104	SMIT	BSc CS	68.0	88.0
4	105	URVISH	BSc IT	45.0	70.0

```
print(df.tail())
```

	Roll No	Name	Department	Marks	Attendance
1	102	DHARMI	BTECH	80.0	90.0
2	103	HEMISH	BCA	88.0	92.0
3	104	SMIT	BSc CS	68.0	88.0
4	105	URVISH	BSc IT	45.0	70.0
5	106	NaN	NaN	NaN	NaN

6b. Write a Python program to analyze the dataset by displaying the number of rows and columns, column names, data types, and basic statistical information. **bold text bold text**

```
print(df.shape) #number of rows and columns
```

```
(6, 5)
```

6c. Write a Python program to filter records from the dataset using conditions. For example, display students having marks greater than 60 or employees having salary greater than 40,000. **bold text**

```
result = df[df["Attendance"]>=60]
print(result)
```

	Roll No	Name	Department	Marks	Attendance
0	101	DIVYESH	BCA	78.0	99.0
1	102	DHARMI	BTECH	80.0	90.0
2	103	HEMISH	BCA	88.0	92.0
3	104	SMIT	BSc CS	68.0	88.0
4	105	URVISH	BSc IT	45.0	70.0

6d. Write a Python program to sort the dataset based on a selected column and display the top 5 records. For example, sort students by marks or products by price. **bold text**

```
result = df.sort_values("Marks")
print(result)
```

	Roll No	Name	Department	Marks	Attendance
4	105	URVISH	BSc IT	45.0	70.0
3	104	SMIT	BSc CS	68.0	88.0
0	101	DIVYESH	BCA	78.0	99.0
1	102	DHARMI	BTECH	80.0	90.0
2	103	HEMISH	BCA	88.0	92.0
5	106	NaN	NaN	NaN	NaN

6e. Write a Python program to perform data analysis on the dataset by finding the maximum, minimum, average, and total values of a numerical column. **bold text**

```
print("Maximum Marks ",df["Marks"].max())
print("Minimum Marks ",df["Marks"].min())
```

```
Maximum Marks  88.0
Minimum Marks  45.0
```